

Music-Gen v4 — Cycles 47-49

2026-09-05

Contents

Music-Gen v4 — Cycles 47-49	1
Abstract	1
Introduction	2
Approach	2
Findings	3
Discussion	5
Open questions	6
Appendix: Provenance	7

Music-Gen v4 — Cycles 47-49

Abstract

Cycles 47-49 extended the preservation-only heartbeat cadence established at the end of the prior range into a mature multi-cycle pattern with a stable, mechanically-repeatable shape while all six operator-authority escalations remained `blocked_on_operator=true` throughout. The range constitutes cycles six, seven, and eight of the consecutive substantive-heartbeat sequence codified by the c36 auditor's M-1/M-2 terminal contracts. Each cycle re-probed the workspace for the `long_exposure/` package (ABSENT across the range — chain-stable since c34), emitted a chain-supersede preservation event on the workspace-forced OPT_B ledger-emitter exemption via string `supersedes_path` per the c14 lemma, scanned `live_guidance` for an operator-supplied c31/c32 POR snapshot (absent across the range), emitted a stand-pat preservation event chain-superseding the prior cycle's stand-pat under the same string-typed supersede semantics, verified the six escalation sidecars byte-identical versus the prior cycle's attestation SHAs, landed four honest-deferral rows for the queued Track B/C/D non-CG bass and drums work with concrete resume commands, verified the Peach Dream stem manifest byte-identical (`c4944ee80dfe446b...`) with the invariant-(d) divergence disclosure carried on the deferral row, extended cross-cycle regression coverage in-place by two cases per cycle, and closed with the standard housekeeping triad (`_plan/register-cN-sub-leaves + _run/cycle_N_closed + _archive/cycle-N-scratch + _infra/adopt-cycleN-tests`) plus emitter-sentinel guard. The chain-supersede lineage now runs `c34 → c35 → c36 → c37 → c38 → c39` (six cycles); the stand-pat lineage now runs `c36 → c37 → c38 → c39` (four cycles). POR parseable-Milestones grew by twelve rows per cycle in a metronomically stable arithmetic — 796 (post-c36) → 808 (post-c37) → 820 (post-c38) → 832 (post-c39) — attributable structurally to the four Track B/C/D honest-deferral rows plus the four-row housekeeping tail plus the two preservation events plus the two test-extension rows, matching the c34-established Track B/C/D deferral-row attribution. Cross-cycle regression coverage grew in-place from twenty-two to twenty-eight cases on `tests/test_c30_legacy_mode_regression.py` (two new tests per cycle capturing the new preservation and stand-pat events' shape and chain-integrity), with the standalone eight-case OP-1 serial-lock suite unchanged — cross-cycle total 36/36 PASS at range close. The c34 empirical POR delta proof diagnostic (SHA `3b0e4d95061a8ad7...`), the c35 blocker event (SHA

c671a40b53565e4e...), the c37 stand-pat (SHA bf701f05d73292a1...), and the c38 stand-pat (SHA 678307e2a59e85c9...) all remained byte-identical across the range. All six operator escalations remained preserved verbatim with correct `carried_from_cycle` semantics (7, 16, 30, 31, 31, 32); no seventh escalation opened; no wait-on-operator memo emitted (banned per operator directive 2026-09-03 part 2). Independent audit at range close returned **VALIDATED** against all eight validators (a)–(h), with zero CRITICAL, zero HIGH, one MINOR-free MODERATE tracking the +12/cycle POR growth trajectory as informational-and-recurring, and a recommendation that the next cycle propose a POR consolidation strategy if operator adjudication of `_manager/M-V4-CERT-composite-fp-drift-adjudication-c32` remains absent. Canonical 7-key `env_pin_sha256=2ac444c3...922ca` unchanged across the range; FD-16(a) re-issue not triggered. Thirteen read-only anchors byte-identical pre-vs-post at range close. `M-V4-SHOWCASE-1` status unchanged (`LANDS_pending_operator`; the CG A/B mix WAV holds SHA 6e13e007... byte-identical since c17). The heartbeat cadence is now demonstrably stable across an extended horizon — six chain-supersede cycles and four stand-pat cycles without degradation, hidden-state accretion, or silent drift.

Introduction

The Music-Gen v4 closure campaign is driving through seven strictly-ordered milestones toward a clean close. Cycles 47-49 sit at what has now become the campaign’s terminal operational cadence: all substantive stage-2 sweep work is gated on operator adjudication of a single composite-floating-point-drift question that spans all three fine-fit drivers (drums c30, bass v2 c31, guitar c31), plus a non-CG bass acceptance-policy escalation from c7, plus a metric-semantics escalation from c16, and no adjudication has arrived through `live_guidance` across the current range or the two preceding ones. Under the anti-heartbeat rule (banned wait-on-operator memos per operator directive 2026-09-03 part 2), the correct posture is preservation-only work that maintains the campaign in a state from which any incoming operator decision can trigger substantive execution in a single downstream cycle.

Two side questions inherited from earlier cycles continue to shape each cycle in the range: the workspace-forced `OPT_B` ledger-emitter exemption established at c34, which requires each cycle to re-probe the workspace and chain-supersede the prior preservation event under the c14 string-typed `supersedes_path` lemma; and the POR shadow-drift strengthening question, which requires each cycle to scan `live_guidance` for an operator-supplied c31/c32 POR snapshot and, absent one, stand-pat via a preservation event chain-superseding the prior cycle’s stand-pat. Both are terminal-contract patterns — the prior auditor’s M-1 and M-2 explicitly directed subsequent cycles to continue mechanically absent a state change. The current range demonstrates the pattern is stable at scale.

Approach

Each of the three cycles executed the same eight-priority template, differing only in the specific SHA anchors chain-superseded and the specific POR arithmetic:

- **Priority 0 (preserve six escalations).** Verify all six escalation sidecars under `data/v4/_manager/` byte-identical versus the prior cycle’s attestation SHAs. No memo mutated. No seventh escalation opened. Anti-stall rule respected — `_manager/` is exhaustive at six.
- **Priority 1 (chain-continuation re-probe).** Execute `test -d long_exposure`, confirm ABSENT (unchanged since c34), emit `data/v4/_selection/cN-emitter-writer-boundary-preservation.json` chain-superseding the prior cycle’s preservation via string `supersedes_path` per the c14 lemma. Confirm `docs/emitter_exemption_policy.md` (SHA fd2c33a78d147341...) byte-identical.
- **Priority 2 (POR stand-pat continuation).** Scan `live_guidance` for operator-

supplied c31/c32 POR snapshot, confirm absent, emit data/v4/_selection/cN-por-drift-preservation.json chain-superseding the prior cycle's stand-pat via string supersedes_path. Confirm c34 empirical proof diagnostic and c35 blocker byte-identical anchors; c34 attribution to Track B/C/D honest-deferral rows transitively holds.

- **Priority 3 (composite-FP-drift adjudication).** Scan live_guidance for PATH_A / PATH_B / PATH_C adjudication token, confirm absent, correctly skip. _manager/M-V4-CERT-composite-fp-drift-adjudication-c32 remains open.
- **Priority 4 (Peach Dream stem manifest anchor).** Verify SHA c4944ee80dfe446b... byte-identical pre-vs-post; carry the invariant-(d) divergence disclosure on the operator_section_c25_checkpointed/rc9_6stem/ non-standard path on the deferral row; no premature disclosure event opened.
- **Priority 5 (honest-deferral rows for Track B/C/D).** Emit four rows with concrete resume commands: Disco A bass stage-2 → next-cycle+ (contingent on Priority 3 + OP-1 SerialLock); Rome bass stage-2 → next-cycle+ (c23 embedding distance 0.5145 predicts SF2_RULED_OUT); Peach Dream bass stage-2 → next-cycle+ (c23 distance 0.4437 + invariant (d) disclosure); WIG + Disco A drums stage-1 → next-cycle+ (coarse-sweep driver green c30; additive --song-sha16 kwarg per c28).
- **Priority 6 (POR shadow-zone hold verification).** Verify parseable_milestones at cycle open matches prior cycle close baseline (delta zero); post-registration count grows by exactly the number of ledger rows emitted this cycle (12 events per cycle across the range). POR baseline 745 (c31 counting-method drift baseline) maintained.
- **Priority 7 (Track F test-suite extension).** Extend tests/test_c30_legacy_mode_regression.py in-place by two cases per cycle (one pinning the cycle's chain-supersede preservation event shape + string supersedes_path semantics; one pinning the cycle's stand-pat event + full chain-integrity through predecessors + c34 diagnostic byte-identity). Standalone tests/test_fine_fit_serial_lock_c32.py unchanged at 8/8.

Housekeeping tail closes each cycle with the standard _plan/register-cN-sub-leaves + _run/cycle_N_closed + _archive/cycle-N-scratch + _infra/adopt-cycleN-tests triad, emitter-sentinel tools/.cN_ledger_emitted guarding against duplicate append.

Discipline guards asserted across all three cycles. All AST-scannable invariants pass: no PRNG imports (random.*, np.random.*), no sidecar_nonfactor, no VST3 state APIs (get_state/save_state/save_preset/load_state/set_state), no --verify-det bypass, /usr/bin/python3 interpreter guard on each cycle's emitter. supersedes_path typed as string (never list) per the c14 lemma throughout. Canonical 7-key env_pin_sha256=2ac44c36298d6ada0579aba1a9160a5881703a4e628f5cccd828b842a922ca unchanged; FD-16(a) re-issue not triggered. FD-16(c) replay-proof invariant carried; no new renders in the range. OP-1 serial-lock invariant remains in force. Zero SF2_CONFIRMED verdicts on non-CG bass (invariant 9 FORBIDDEN under SHOWCASE-1-non-cg-bass-acceptance-policy operator authority). No wait-on-operator memo emitted; Priority 0 preserve-only posture satisfies the genuine operator-authority carve-out.

Findings

Chain-supersede lineage matured to six cycles

The c34-established workspace-forced OPT_B ledger-emitter exemption chain now runs six cycles deep: c34 fork → c35 preservation → c36 preservation → c37 preservation → c38 preservation → c39 preservation. Each cycle's preservation event carries supersedes_path as a string pointing at the immediately-prior cycle's preservation event path, exercising the c14 lemma each cycle rather than treating it as dormant convention. docs/emitter_exemption_policy.md (SHA fd2c33a78d147341...) remains byte-identical anchor throughout. The reversibility contract remains intact: if any future cycle finds long_exposure/ PRESENT, an OPT_A adjudication event

can supersede the latest preservation and route the emitter chain through `append_ledger_event`.

Stand-pat lineage matured to four cycles

The POR shadow-drift strengthening stand-pat pattern established at c36 now runs four cycles deep: c36 stand-pat → c37 stand-pat → c38 stand-pat → c39 stand-pat. c37 stand-pat (SHA bf701f05d73292a1...), c38 stand-pat (SHA 678307e2a59e85c9...), c35 blocker (SHA c671a40b53565e4e...), and c34 empirical proof diagnostic (SHA 3b0e4d95061a8ad7...) all remained byte-identical across the range. The c34 attribution of the c31→c32 +4 shadow-drift to the four Track B/C/D honest-deferral rows continues to hold transitively per string-supersede semantics.

Six operator escalations preserved verbatim across the range

All six escalation sidecars under `data/v4/_manager/` verified byte-identical versus each cycle's prior attestation:

Escalation	Origin	Attestation SHA
SHOWCASE-1-non-cg-bass-acceptance-policy	c7	8101f7d57ef52991...
M-V4-METRIC-SEMANTICS-c16	c16	011a708e94989e6a...
CERT-fine-fit-sf2-drums-legacy-halt	c30	aeaafabfadd4d83d...
CERT-fine-fit-sf2-v2-legacy-halt	c31 (c33 sidecar backfill)	4b95efe95c551b0a...
CERT-fine-fit-sf2-guitar-legacy-halt	c31 (c33 sidecar backfill)	108b48af93a88548...
CERT-composite-fp-drift-adjudication-c32	c32	c4735de75895b46e...

The c32 composite-FP-drift consolidation memo links but does not close the three per-driver fine-fit HALT predecessors, preserving FD-1 discipline across the c30 / c31 / c31 / c32 chain. All six carry correct `carried_from_cycle` semantics; all `blocked_on_operator=true`; all discoverable via both `grep _manager/M-V4-CERT` and `ls data/v4/_manager/`. Priority 0 status `BLOCKED_ON_OPERATOR` unchanged across the range.

POR parseable-Milestones +12/cycle heartbeat now four data points

The POR parseable-Milestones count now traces a metronomically stable +12/cycle arithmetic across the four consecutive heartbeat data points at range boundaries: 796 (post-c36) → 808 (post-c37) → 820 (post-c38) → 832 (post-c39). Each cycle open matches the immediately-prior cycle close exactly (delta zero); each post-registration count grows by exactly twelve rows from twelve emitted ledger events (four Track B/C/D deferral rows + four housekeeping-tail rows + two preservation events + two test-extension rows, matching the c34-established Track B/C/D deferral-row attribution). POR baseline 745 (c31 counting-method drift baseline) maintained through c32 / c33 / c34 / c35 / c36 / c37 / c38 / c39. No new drift beyond the accounted heartbeat.

Read-only anchors held throughout the range

Thirteen read-only anchors verified byte-identical pre-vs-post at range close: `scripts/sound_match/objective.py` 8087ce80...; `scripts/sound_match/_sweep_hygiene_c27.py` 771ff42b...; `scripts/sound_`

match/_serial_lock_op1.py 121809db...; data/v4/deliveries/31a164f845f8e27e/cg_ab_mix.wav 6e13e0075c5d8116...; docs/agent_picks_selection_invariants.md post-c32 29a1610b...; docs/emitter_exemption_policy.md fd2c33a7...; three fine-fit drivers post-OP-1 (6c80c438, a432e1d1, 40dbb673); three coarse-sweep drivers (3f8bfa08, 26aa754c, d6c54f21); Peach Dream stem manifest c4944ee8...; plus the six escalation memos under data/v4/_manager/.

Zero SF2_CONFIRMED verdicts anywhere on disk. Canonical 7-key environment pin unchanged.

Test coverage grew 22 → 24 → 26 → 28 in-place

Cross-cycle regression coverage on tests/test_c30_legacy_mode_regression.py extended in-place by two cases per cycle across the range, capturing each cycle's new preservation and stand-pat events' shape and chain-integrity. Growth trajectory: c36 22 → c37 24 → c38 26 → c39 28. Standalone tests/test_fine_fit_serial_lock_c32.py unchanged at 8/8. Cross-cycle total at range close: 36/36 PASS.

Audit outcome

VALIDATED at range close against all eight validators (a)–(h). Zero CRITICAL. Zero HIGH. One MODERATE, informational-and-recurring, tracking the +12/cycle POR growth trajectory as structurally honest per c34-established Track B/C/D deferral-row attribution plus housekeeping-tail arithmetic — not substantive accretion. The prior audit projected that consolidation-strategy discussion might merit consideration at c40+; the current audit continues the not-action-required ruling per the original scope of the observation and recommends that the next cycle propose a consolidation strategy (parallel to c31/c34 shadow-zone hygiene precedent) if operator adjudication of _manager/M-V4-CERT-composite-fp-drift-adjudication-c32 remains absent, retaining the current mechanical pattern in the meantime.

Zero MINOR observations at range close. The audit interprets the range as a textbook instance of the halt-honest campaign closure discipline the brief mandates — five consecutive Priority 1 chain-supersede preservations and four consecutive Priority 2 stand-pat continuations demonstrate the mechanical heartbeat pattern is genuinely stable across an extended horizon, not degrading, not accreting hidden state, not silently drifting.

Discussion

Three things about this range are worth naming.

First, the range is the empirical demonstration that the preservation-only cadence established at the end of the prior range is not a transient posture but a stable multi-cycle equilibrium. Six chain-supersede cycles and four stand-pat cycles now stand as evidence that the pattern can continue mechanically without degradation: each cycle's disk-state matches the prior cycle's expected close-state exactly, each cycle's ledger emission is a clean twelve rows, each cycle's test extension is a clean two cases, each cycle's read-only anchor set is byte-identical. There is no shear stress accumulating between cycles. This is what "hold cleanly while blocked on operator" looks like at scale — it is not idling and it is not heartbeat-manufacturing. Each cycle re-executes the discipline invariants (workspace probe, supersedes_path string-typing, escalation byte-identity attestation, POR arithmetic), keeps the campaign reversible-ready, and closes.

Second, the +12/cycle POR growth trajectory is now sharp enough to be predictable and structurally attributable. Every twelve-row growth per cycle decomposes exactly to four Track B/C/D deferral rows + four housekeeping-tail rows + two preservation events + two test-extension rows. The c34 empirical proof anchored this attribution; three subsequent cycles

of the same arithmetic without drift is confirmatory evidence. This is why the audit continues to characterize the growth as “informational, non-blocking” rather than substantive accretion — the delta per cycle is fully accounted-for at row-level, and the accumulation is the natural consequence of running an operator-authority hold under the anti-heartbeat rule. If operator adjudication of the composite-FP-drift memo remains absent through a subsequent cycle, however, the growing row count merits a consolidation-strategy proposal parallel to earlier shadow-zone hygiene work — not to change the substance of what is being tracked, but to compact the accumulated deferral-row set into a stable summary that the campaign can hand off cleanly at whichever cycle operator adjudication arrives.

Third, the c14 string-typed `supersedes_path` lemma has now been exercised across ten preservation events in ten cycles (six chain-supersedes on the exemption question + four stand-pats on the POR question) without a single lapse. Each cycle re-executes the lemma against a live prior-cycle anchor; each cycle’s audit re-attests the string type. The lemma is not dormant convention; it is living infrastructure that makes the preservation-only cadence auditable and reversible. When operator adjudication does arrive — likely Path A (accept render-level regression bar via new invariant (f)) as the single-action resolution across all three fine-fit drivers — the transition from this stable state to substantive stage-2 sweep launch requires only that the adjudication token appear in `live_guidance`; the next cycle’s worker executes the chosen path and the deferral chain collapses. The range’s investment in maintaining that clean state is what keeps the transition cheap.

Open questions

- **Composite-FP-drift operator adjudication.** The three per-driver fine-fit legacy HALT escalations (drums c30, bass v2 c31, guitar c31) plus the c32 consolidation memo constitute a single operator-authority question with three named paths (A accept render-level bar via new invariant (f); B hold strict; C harden `objective.py` per invariant 8 operator scope). Path A is the single-action resolution across all three drivers. No adjudication in `live_guidance` across the range.
- **Non-CG bass acceptance-policy escalation.** Remains `blocked_on_operator=true`. `SF2_CONFIRMED` remains FORBIDDEN on non-CG bass. Systematic 4-arc composite-vs-source-of-truth finding still predicts OPT2 (refuse extension + OPT3 `htdemucs` bass fallback) as the invariant-compliant outcome.
- **Metric-semantics escalation.** Remains `blocked_on_operator=true` with `carried_from_cycle=16`.
- **Emitter writer-boundary chain.** Six-cycle chain-supersede continues if `long_exposure/` remains ABSENT. Reversibility contract intact — if a future cycle finds it PRESENT, OPT_A adjudication event supersedes the latest preservation.
- **POR shadow-drift stand-pat.** Four-cycle chain continues absent an operator-supplied c31/c32 POR snapshot. c34 empirical proof transitively holds.
- **POR consolidation strategy proposal (auditor recommendation for next cycle).** With four consecutive +12/cycle data points and no operator adjudication imminent, the next cycle’s worker is asked to propose a consolidation strategy parallel to c31 / c34 shadow-zone hygiene precedent, retaining the current mechanical pattern in the meantime. This would compact the accumulated Track B/C/D deferral-row set into a stable summary that hands off cleanly at whichever cycle operator adjudication arrives.
- **Expected next-cycle open state.** `parseable_milestones` = 832; six escalation memo SHAs unchanged; `long_exposure/` ABSENT; c39 preservation + stand-pat + POR-hold events on disk byte-identical; 36/36 test suite green pre-extension.
- **Two operator triggers that would change the state.** Either (a) an operator-supplied c31/c32 POR snapshot (would fire a Priority 2 close event superseding the stand-pat

chain), or (b) a PATH_A / PATH_B / PATH_C adjudication token for the composite-FP-drift memo (would unblock Priority 3 downstream + Priority 4 Peach Dream disclosure + Priority 5 non-CG bass and drums sweep resumption). Both remain absent.

Appendix: Provenance

Directive. Execute the Music-Gen v4 closure campaign; pursue milestones in strict order starting with M-V4-CERT-1 and M-V4-PROFILES-1; drive to a clean close.

Cycle range. cycles 47-49.

Working directory. /home/user/long-exposure-runs/music-gen.

Session references.

- Cycle 47 researcher 4f964648-4b5a-4134-917f-68e940f4639a; worker e0f14848-e752-418e-a79b-670b9777b225; auditor 24c24104-9c9a-4ce0-ae1e-338f62e1f378.
- Cycle 48 researcher 31c7cd68-de31-4643-a12e-9a016469a04c; worker c21aa847-f6d7-4b32-948e-a1279d07d81a; auditor 2b5059d4-60fc-448a-9bed-d3a157c8f869.
- Cycle 49 researcher 58cbaffa-1d20-4688-b324-bd6658b6e9df; worker a967b16c-3b0f-483c-9d99-f6126f4af19a; auditor b9b28d40-7f1b-44f8-b3bd-12f1f76afa38.

Audit verdict. VALIDATED at range close against all eight validators (a)-(h). Zero CRITICAL. Zero HIGH. One MODERATE (informational, non-blocking, recurring): POR parseable-Milestones +12/cycle heartbeat now four data points (796 → 808 → 820 → 832); structurally honest per c34-established Track B/C/D deferral-row attribution + housekeeping-tail arithmetic; recommendation that next cycle propose consolidation-strategy parallel to c31/c34 shadow-zone hygiene precedent if operator adjudication remains absent. Zero MINOR.

Terminal deliverables landed this range.

- Three chain-supersede preservation events on the workspace-forced OPT_B ledger-emitter exemption at data/v4/_selection/c{37,38,39}-emitter-writer-boundary-preservation.json, each supersedes_path string-typed → prior-cycle preservation per c14 lemma.
- Three stand-pat preservation events on the POR shadow-drift strengthening question at data/v4/_selection/c{37,38,39}-por-drift-preservation.json, each supersedes_path string-typed → prior-cycle stand-pat.
- Twelve honest-deferral rows total across the range for Track B/C/D (four per cycle: Disco A bass stage-2; Rome bass stage-2; Peach Dream bass stage-2 with invariant-(d) disclosure carried; WIG + Disco A drums stage-1) with concrete resume commands.
- Six new test cases across the range (two per cycle) extending tests/test_c30_legacy_mode_regression.py in-place 22 → 24 → 26 → 28; standalone tests/test_fine_fit_serial_lock_c32.py unchanged at 8/8; cross-cycle total 36/36 PASS at range close.
- Three housekeeping-triad sequences (one per cycle): _plan/register-cN-sub-leaves + _run/cycle_N_closed + _archive/cycle-N-scratch + _infra/adopt-cycleN-tests; emitter-sentinel guards tools/.c{37,38,39}_ledger_emitted.

Ledger + POR arithmetic across the range.

- c37: ledger 1600 → 1612 (+12); POR 796 → 808 (+12).
- c38: ledger 1612 → 1624 (+12); POR 808 → 820 (+12).
- c39: ledger 1624 → 1636 (+12); POR 820 → 832 (+12).
- Each cycle open matches prior cycle close (delta zero). POR baseline 745 (c31 counting-method drift baseline) maintained.

Read-only anchors preserved byte-identical pre-vs-post (13 verified at range close).

- scripts/sound_match/objective.py 8087ce80... (invariant 8; Path C in HALT escalations would require lifting)
- scripts/sound_match/_sweep_hygiene_c27.py 771ff42b...
- scripts/sound_match/_serial_lock_op1.py 121809db...
- data/v4/deliveries/31a164f845f8e27e/cg_ab_mix.wav 6e13e0075c5d8116...
- docs/agent_picks_selection_invariants.md post-c32 29a1610b...
- docs/emitter_exemption_policy.md fd2c33a7...
- Three fine-fit drivers post-OP-1: 6c80c438, a432e1d1, 40dbb673
- Three coarse-sweep drivers: 3f8bfa08, 26aa754c, d6c54f21
- Peach Dream stem manifest c4944ee8...
- Six escalation memos under data/v4/_manager/ (as tabulated in Findings)

Chain-supersede provenance anchors verified byte-identical across the range.

- c34 empirical POR delta proof diagnostic 3b0e4d95061a8ad7...
- c35 blocker event c671a40b53565e4e...
- c37 stand-pat bf701f05d73292a1...
- c38 stand-pat 678307e2a59e85c9...

Six operator escalations preserved verbatim with correct `carried_from_cycle` semantics (7, 16, 30, 31, 31, 32); all `blocked_on_operator=true`; all discoverable via both `grep _manager/M-V4-CERT` and `ls data/v4/_manager/`. Priority 0 status `BLOCKED_ON_OPERATOR` unchanged across the range.

Environment pin. Canonical 7-key `env_pin_sha256=2ac444c36298d6ada0579aba1a9160a5881703a4e628f5ccdd828b842` unchanged this range; FD-16(a) re-issue not triggered. FD-16(c) replay-proof ×2 per render family per song invariant carried; no new renders in the range.

Discipline guards asserted (AST-scannable). No PRNG imports (`random.* / np.random.*`), no `sidacar_nonfactor`, no VST3 state APIs, no `--verify-det` bypass, `/usr/bin/python3` interpreter guard on each cycle's emitter. `supersedes_path` typed as string (never list) per c14 lemma across all ten preservation events emitted this range. Zero `SF2_CONFIRMED` verdicts on non-CG bass (invariant 9 FORBIDDEN under `SHOWCASE-1-non-cg-bass-acceptance-policy` operator authority). OP-1 serial-lock invariant in force. No wait-on-operator memo (banned per operator directive 2026-09-03 part 2); Priority 0 preserve-only posture satisfies genuine operator authority carve-out.

Milestone status at range close.

- M-V4-CERT-1 — validated (E2E_DETERMINISM_HOLDS on the v3 spine).
- M-V4-PROFILES-1 CG (5/5 instruments) — validated.
- M-V4-PROFILES-1 non-CG bass — 2/4 `SF2_RULED_OUT` (Rome, Peach Dream); 2/4 `STILL_INDETERMINATE` (WIG, Disco A). Stage-2 re-work gated on operator adjudication.
- M-V4-PROFILES-1 non-CG drums — 0/4 (Track D unblocked at driver-regression level; stage-2 gated on operator adjudication).
- M-V4-PROFILES-1 non-CG guitar — 0/2 (WIG + Peach Dream guitar are NULL by earlier MIDI-probe).
- M-V4-SHOWCASE-1 CG — `LANDS_pending_operator` (`cg_ab_mix.wav` SHA 6e13e007... byte-identical since c17).
- M-V4-SHOWCASE-1 non-CG — `BLOCKED` on non-CG bass acceptance-policy escalation.
- M-V4-RULES-1 — scaffold landed c20; substantive implementation queued.
- M-V4-EAR-1 — not yet opened.
- M-V4-GEN-1 — conditional on M-V4-RULES + M-V4-EAR.
- M-V4-CLOSE-1 — c24 amendment landed; further amendments as substantive work completes.

Next-cycle first task. Continue the preservation-only cadence per c39 auditor forward guidance: Priority 1 chain-continuation (re-probe `long_exposure/`; chain-supersede c39 preservation if still ABSENT); Priority 2 stand-pat continuation (scan `live_guidance` for operator-supplied POR snapshot; stand-pat if absent); Priority 3 composite-FP-drift adjudication contingent on operator; Priorities 4-6 as usual; Priority 7 test extension by two cases. **Consider proposing a POR consolidation strategy** per c39 MODERATE M-1 parallel to c31/c34 shadow-zone hygiene precedent if the operator-adjudication wait continues. If any operator adjudication arrives via `live_guidance` — either a c31/c32 POR snapshot for Priority 2 close, or `PATH_A/PATH_B/PATH_C` for the composite-FP-drift memo — execute the chosen path (Path A recommended as single-action resolution across all three fine-fit drivers). Operator ear remains LANDS authority post-hoc per FD-6.